DIC22
Four-to-One Multiplexer Comparison

Specification-based laboratory report

17 October 2026

English edition
HSPICE S-2021.09 | Model library supplied separately

1 Objectives

Implement the Lab 22 four-to-one MUX from PDF pages 88–90, including both handout architectures, a faster TG design, and delay/power versus load.

2 Specification Mapping

S denotes simulated checks; D denotes analytical discussion.

Requirement and PDF pages	Deck / check	Status
Fig.22.3 TG/static CMOS, 10fF (88–89)	D01; D02/C0	S
Delay, power and device counts (89)	D02; source probes	S+D
Design a faster TG MUX (89)	Wide unbuffered tree	S; data path
Load dependence of all three MUXes (90)	D02; 1–500fF	S
Complete 4-to-1 truth table	D01; 64 codes	S

3 Circuit and Design Derivation

$$Y = \overline{S}_1\overline{S}_0A_0 + \overline{S}_1S_0A_1 + S_1\overline{S}_0A_2 + S_1S_0A_3.$$

The lecture TG circuit uses an input inverter and two series TGs per path, followed by one shared output inverter. Static CMOS uses four NAND3 gates and one NAND4. The new design is a two-level binary tree with six TGs, each at eight times unit width, directly driving the output without output inverters. All L are 0.25 μm, with unit $W_p/W_n = 3.6/1.8\,\mu$m; stacked NAND NMOS widths scale with stack length. Each architecture adds two select inverters. The new tree has no gain restoration and depends on upstream data drive; fewer devices do not imply smaller total area.

Zero-volt probes at every data input separately measure input energy, preventing supply-only power from hiding the unbuffered tree's cost. Supply+data still excludes ideal select-source energy and does not model a real upstream driver.

4 Simulation Setup

The model is supplied separately on the user's computer. These are 1 V, 27 °C teaching conditions; the handout's old-process 2.5 V numbers are not pass thresholds. Actual PrimeSim HSPICE S-2021.09 runs use `accurate`, `runlvl=5` and `.print/.measure` data. Every `.alter` resets baseline parameters and temperature. All cases completed without errors or failed measurements; model and convergence notices may remain. Public netlists retain only `* MODEL_SETUP_REQUIRED`, with no model path. Full data are under the lab's `results/`; large files are split by deck and indexed by `simulation_data.json`.

Two decks contain six cases. D01 enumerates all 64 data/select combinations, 2 ns each, checking outputs at 80% of each slot against 0.1/0.9 V thresholds. D02 holds select=00, drives A0 with 20 ps edges and a 10 ns period, and loads 1/10/50/100/500 fF, with at most 1 ps steps. Delay uses A0-to-output 50% crossings; this is not worst-case select delay.

5 Results and Specification Checks

Architecture	MOS core/all	t_{rise} (ps)	t_{fall} (ps)
Lecture TG	26/30	381.86	560.61
Static CMOS	32/36	230.90	364.48
Wide TG tree	12/16	52.16	92.85

Architecture	P_{supply} (μW)	P_{data} (μW)	P_{sum} (μW)
Lecture TG	23.886	2.943	26.829
Static CMOS	19.060	2.311	21.371
Wide TG tree	-1.720	66.417	64.698

Load (fF)	TG max (ps)	CMOS max (ps)	Tree max (ps)
10	560.61	364.48	92.85
1	507.14	348.69	90.62
50	692.41	413.43	103.46
100	799.99	474.38	116.16
500	1586.69	1426.36	208.96

All three architectures pass all 64 truth-table combinations. At 10 fF, the new tree gives 52.16/92.85 ps rise/fall delay, below static CMOS at 230.90/364.48 ps, meeting the selected data-path speed goal. Yet supply+data power for the truth stimulus is 64.70 μW, above CMOS at 21.37 μW. The tree's -1.72 μW supply-only value is not an energy-saving result.

6 Discussion

Wider TGs lower channel resistance and removing inverters saves gate delay, but increases data/select capacitance and shifts output charging energy to the upstream source. Finite drivers, interconnect or cascaded MUXes can reduce the advantage and require regeneration. Static CMOS restores gain and provides active output drive. The original TG version provides regeneration but adds transmission delay. All delays increase with load; the ordering applies only to these sizes and input conditions.

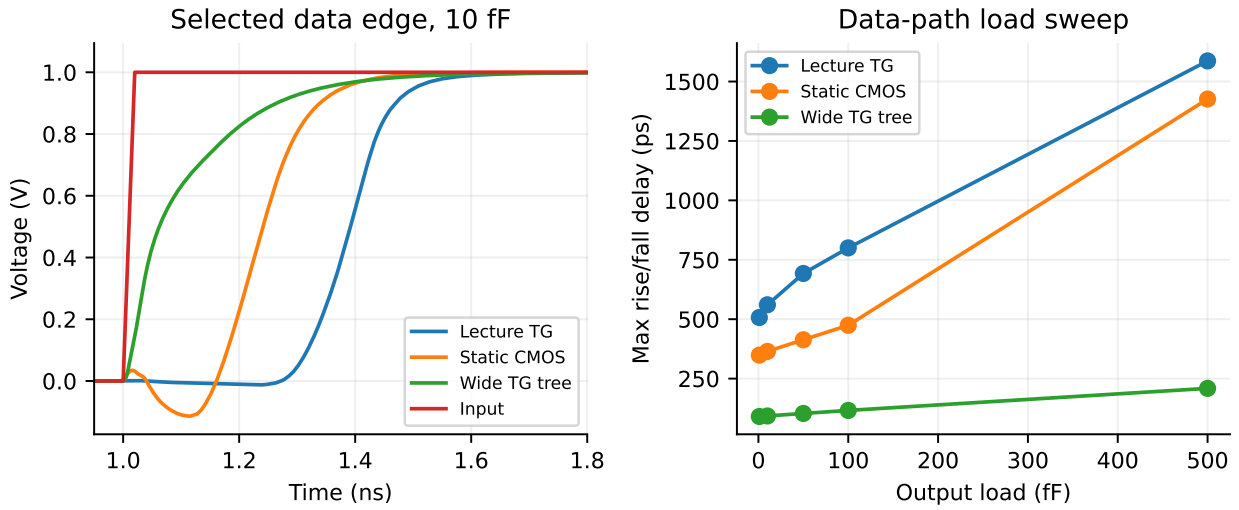


Figure 1: 10 fF data path and load sweep. The faster tree requires the data source to supply output charging current directly.

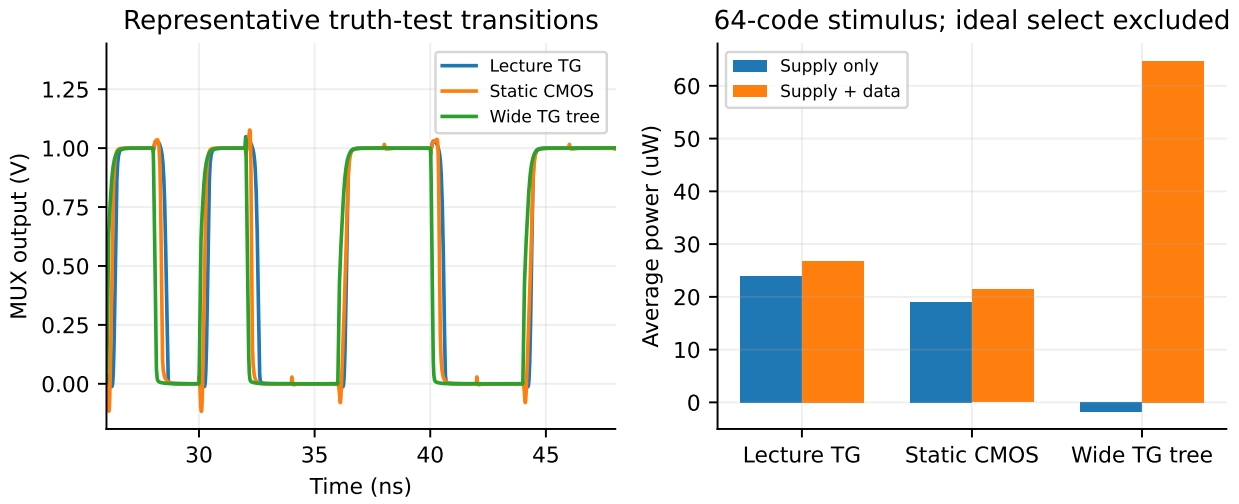


Figure 2: Representative truth-test transitions and 64-code power. Negative supply power is net returned energy, not negative total consumption.

Transition glitches and rail excursions remain visible rather than being hidden by settled truth checks. Counts exclude stimulus, loads and probes; differing widths also prevent transistor counts from representing layout area.

7 Conclusion

The handout circuits, faster TG design, exhaustive truth table and five loads are verified. Faster selected-data timing costs input energy and gain restoration, and is not a universal optimum for all paths or complete systems.

8 MOS transistor schematic

These editable vector schematics connect equal node names within a circuit. Primitive-cell names are local pins; the hierarchy defines their connections. B denotes the body terminal; 0 is global ground. Each panel identifies its netlist and scope; measurement sources are connected as described. Exact cells and hierarchy for `01_mux_truth_table.sp` and `02_mux_delay_load.sp`.

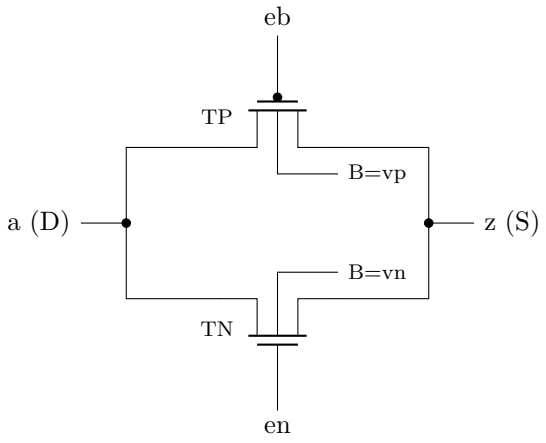


Figure 3: Pass-gate cell used in both TG architectures.

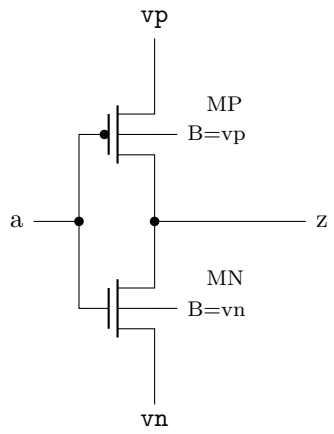


Figure 4: Input/output and local select-complement inverter.

The inv cell has $W_P = 3.6k\,\mu$m, $W_N = 1.8k\,\mu$m and $L = 0.25\,\mu$m. The tg cell uses the same widths/length. For tg, both device drains are a and both sources are z; the pass path is bidirectional. TP/TN correspond to MP/MN inside the tg subcircuit. PMOS bodies remain vp and NMOS bodies vn, independent of the signal terminals. The n-channel gate is en, and the p-channel gate is eb. Equal node labels connect the two pass devices in parallel.

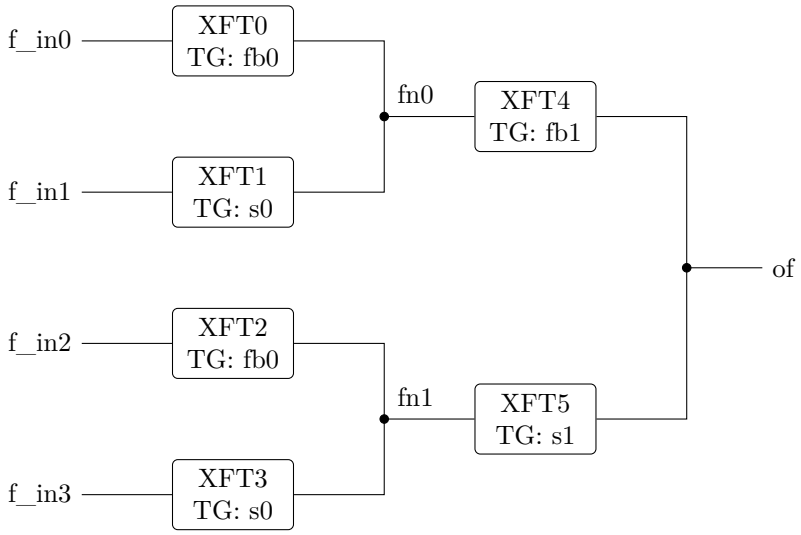


Figure 5: Optimized TG tree: the label after TG is en, eb is its local complement. All six cells use k=8.

fb0/fb1 are generated by two inv cells from s0/s1 using vf/0 rails. Every TG in this tree has body rails vf/0. CLf connects of to 0. No output inverter is present: six TGs plus two select inverters give 16 MOS. The input probes VDfi connect f_ini (positive) to ai (negative) at zero volts. The corresponding g/c input probes use the same convention.

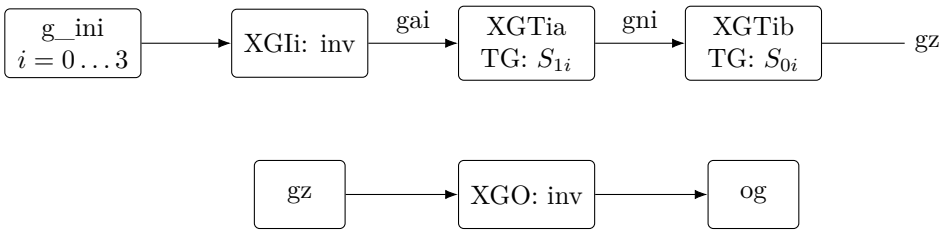


Figure 6: Lecture TG architecture: four identical indexed branches join at gz; a single output inverter regenerates og.

$S_{1i} = s1$ for $i \geq 2$, otherwise gb1; $S_{0i} = s0$ for odd i, otherwise gb0. eb is the opposite select literal. All cells use k=1 and vg/0 rails; two local select inverters are included. The four branch inputs and output each have an inverter, giving 30 MOS including select inversion.

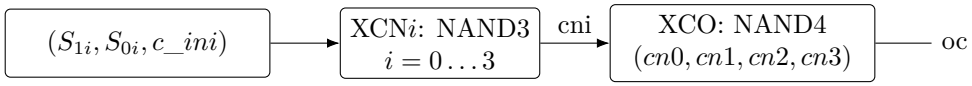


Figure 7: Static CMOS MUX hierarchy; all four NAND3 outputs enter one NAND4.

Here the select complements are cb0/cb1, generated locally with vc/0 rails. All static gates use k=1. Outputs og/oc/of each have the same cl load to 0.

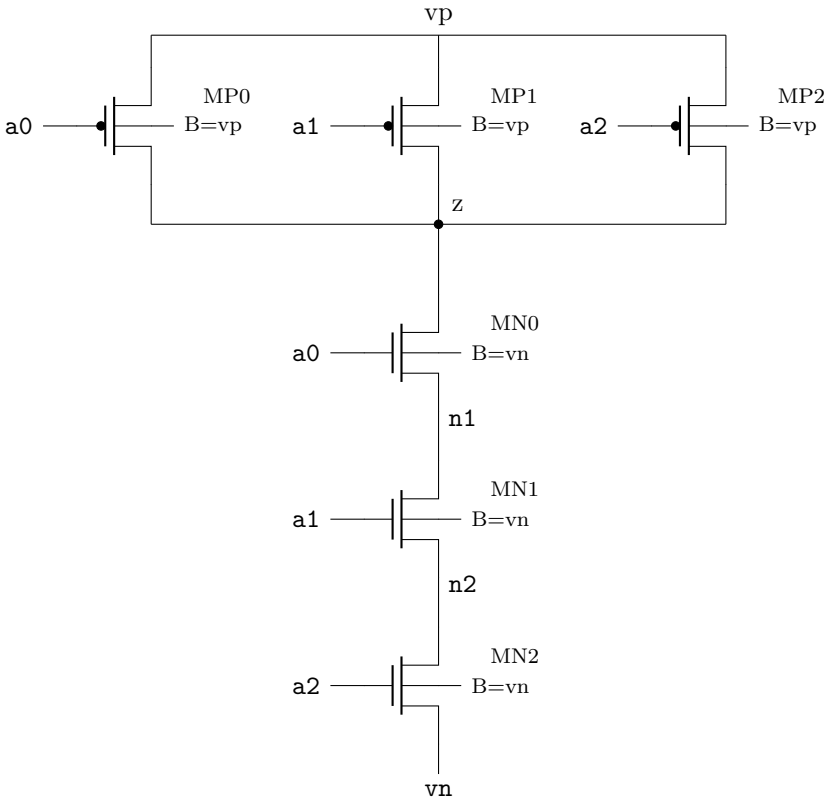


Figure 8: Exact NAND3 transistor cell: three parallel PMOS branches and the n1/n2 NMOS series stack.

For nand3, $W_P = 3.6k\,\mu$m and $W_N = 5.4k\,\mu$m, $L = 0.25\,\mu$m. nand4 extends the same parallel-PMOS/series-NMOS pattern to four inputs, with $W_N = 7.2k\,\mu$m and the same PMOS width and length. All PMOS bodies/sources connect to vp; every NMOS body connects to vn, while intermediate sources are the stack nodes. Thus four NAND3 cells (24 MOS), one NAND4 (8 MOS), and two select inverters (4 MOS) give 36 MOS.